

SEED CLUSTERING WITH K-MEANS

Short Description

In this project I used unsupervised machine learning to group wheat seeds based on their measurements. The aim was to see if K-Means clustering could find the real wheat varieties without using the variety labels.

Problem Statement

The problem was to find out how many groups of wheat seeds were present in the data without using the real variety labels. I also wanted to see how well the groups found by K-Means matched the real wheat varieties afterwards.

Dataset

The Seeds dataset was downloaded from Openml. The seeds dataset contains 210 wheat seeds. Each seed has 7 different measurements such as area, perimeter, compactness, length, width, asymmetry and groove length. The real wheat variety labels were kept hidden at first and only used later to check to see how well the clustering worked.

Method

First the data was scaled so the different measurements could be compared fairly. I then used PCA to reduce the 7 measurements to 2 dimensions so the data could be shown on graph. After that I used the elbow method and silhouette score to help choose the number of clusters. I chose 3 clusters and then used K-Means to group the seeds. Finally I compared the clusters with the real wheat varieties using a cross tabulation and purity score.

Results

The elbow method suggested 3 clusters while the silhouette score was highest at 2. I chose 3 clusters because the elbow graph showed a clear bend at 3 and the silhouette score was still fairly high. After running K-Means, the purity score was about 0.92. The cross-tabulation showed that each cluster was mostly made up of one real wheat variety with 193 out of 210 seeds matching the main variety in their cluster. Variety 3 was separated the most clearly.

Interpretation

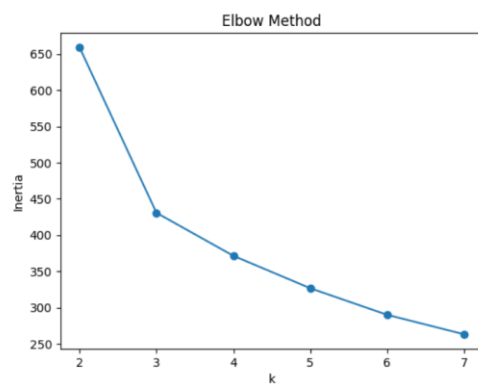
The results show that K-Means was able to find most of the real wheat variety structure without using the labels. A purity score of 0.92 shows that the clusters matched the real varieties really well but not perfect. There was still some overlap between varieties so the clusters should not be treated as completely certain.

Reflection

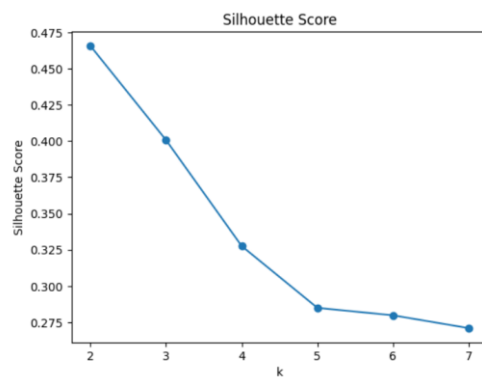
I learned how K-Means can group data without using labels and how the elbow method and silhouette score can help choose the number of clusters. At first I found it difficult to understand why the elbow method and the silhouette score gave different answers but I understood that they measure the clusters in different ways. If I had more time I would try other clustering methods and compare the results with K-Means.

The Results

- Elbow method suggested 3 clusters:



- Silhouette score was highest at 2 but it was fairly high at 3:



- The three varieties were mostly separated even though there was some overlap:

